Unit: Unit 3: Functions
Topic: Function Transformations

Name: ____________
Date: ________

Advanced (Habanero)

Open Ended Questions (FRQ)

Q9. Describe the transformations applied to the graph of $y = f(x)$ to obtain the graph of $y = -2f(x+1) - 3$. Support your answer with $x \cdot y$ coordinates of the vertex of the parabola $y = x^2$.

Q10. Compare the graphs of $y = f(x)$ and $y = f(x-2) + 1$. How do the $x \cdot y$ intercepts and the vertex of the parabola $y = x^2$ change after applying these transformations?

Chef's Tips

- Tip 1: Encourage students to visualize and sketch graphs when applying transformations.
- Tip 2: Remind students that $x \cdot y$ coordinates are essential in describing transformations and that the order of transformations matters.
- Tip 3: Advise students to use graph paper for $x \cdot y$ plotting.